Project Proposal

Temporal Regulatory Knowledge Graph System using Ontology and LLM

Project Code

- To be assigned by the Project Office

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1. Abstract

Government laws, rules, notifications, amendments and policies are spread across many separate documents and keep changing over time. A requirement may be written in an Act, explained in Rules and then changed by a notification with its own starting date. As a result, people find it hard to know *which rule applies today*, *which rule applied on an earlier date*, *what changed* from the previous version, and *which services or requirements are affected*. Cyber law in Pakistan shows this problem clearly: the Prevention of Electronic Crimes Act (PECA), 2016 has been changed by rules, an ordinance and the PECA (Amendment) Act, 2025, which added new offences, new regulatory bodies and a new investigation agency [1]–[3], [18].

In this project we will build a **Temporal Regulatory Knowledge Graph System using Ontology and LLM**, with cyber law as its main working domain. We will: (i) design a general regulatory ontology for laws, amendments, requirements, obligations, departments, services and dates, and extend it with cyber law concepts such as offences, penalties and investigating authorities; (ii) store all versions of each rule in a Knowledge Graph (KG) with the dates they were valid; (iii) detect amendments, find their effective dates and compare old and new versions; (iv) trace the affected services, offences, departments and procedures; and (v) connect a Large Language Model (LLM) through a date-aware Graph-RAG pipeline that answers questions such as "What changed in PECA in 2025?", "Which section applied to an offence committed in 2024?" or "Which services are affected by this amendment?" with simple answers and references to official documents. After cyber law, the same design will be extended to other areas of law.

The project will provide a reusable regulatory ontology, a complete cyber law knowledge graph for Pakistan, a date-aware Graph-RAG method and a test set. It can help citizens, government departments, lawyers, investigators and online platforms understand both current and past rules.

2. Background and Justification

In Pakistan, laws are passed by the Parliament and Provincial Assemblies, rules are made under them by ministries and departments, and changes come through notifications, Statutory Regulatory Orders (SROs), circulars and policies. These are published separately, often as scanned PDFs, so finding the rule for a certain date means collecting every related document and applying each change by hand.

Cyber law is one of the fastest-changing areas and is the main domain of this project. PECA 2016 defines offences such as unauthorised access to information systems and cyber stalking, and section 37 allows the Pakistan Telecommunication Authority (PTA) to remove or block unlawful online content [1]. How PTA should do this was later explained in the Removal and Blocking of Unlawful Online Content (Procedure, Oversight and Safeguards) Rules, 2021 [3]. In February 2022 the PECA (Amendment) Ordinance, 2022 was issued [18], but on 8 April 2022 the Islamabad High Court struck it down and also declared part of section 20 unconstitutional [19]. Then the PECA (Amendment) Act, 2025 (Act No. II of 2025), published on 29 January 2025, added section 26A on spreading false or fake information (up to three years in prison, a fine of up to Rs. 2 million, or both), created the Social Media Protection and Regulatory Authority, a Social Media Complaint

Council and Social Media Protection Tribunals, and set up the National Cyber Crime Investigation Agency (NCCIA) to replace the FIA Cyber Crime Wing [2]. The Electronic Transactions Ordinance, 2002 also applies to electronic documents and signatures [4]. So, for a cyber complaint, the answer to "which agency investigates it?" or "what is the penalty?" depends on the date of the offence, and online platforms must follow rules that keep changing.

What has already been done (2024–2026). Recent work combines knowledge graphs with Retrieval-Augmented Generation (RAG), where an LLM retrieves information before answering: GraphRAG, HippoRAG and LightRAG use graphs to connect facts [5], [8], [9], surveys review graph-based RAG [6], [7], and Pan et al. show that graphs give LLMs reliable facts [10]. Zep stores facts in a temporal knowledge graph that remembers when each fact was true and when it was recorded [11]. The closest work to ours is an ontology-based Graph RAG for legal norms that keeps every version of a provision and treats each amendment as a separate event [12]. GraphCompliance turns the GDPR, a data protection law closely related to cyber law, into a graph and beats plain LLM and RAG methods [13], and Echenim and Joshi used a knowledge graph with an LLM to check IoT privacy and security compliance [14]. LegalBench-RAG offers a benchmark for legal retrieval [15], while a 2025 study found that commercial AI legal tools still gave wrong or made-up answers in more than 17% of cases [16].

Why this project is needed. These studies show that knowledge graphs make LLM answers more accurate and easier to explain, but gaps remain: (i) normal RAG does not consider dates, so it may return an old or repealed rule; (ii) existing work rarely links a rule change to the services, requirements and departments it affects; and (iii) no such system exists for Pakistani cyber law, where users must currently compare PECA, its amendments and PTA rules by hand. Our project fills these gaps with an ontology, a knowledge graph with validity periods, and an LLM that answers only from checked evidence valid on the asked date.

3. Research Methodology

We will follow an iterative Agile approach in short sprints. The first two phases are:

- **Requirement Analysis:** identifying users (citizens, victims of cybercrime, lawyers, investigators, online platforms and departments) and their questions, writing the questions the ontology must answer, and preparing the Software Requirements Specification (SRS).
- **Data Collection and Preparation:** collecting PECA 2016, the 2022 ordinance, the 2025 amendment, the 2021 content-blocking rules, the Electronic Transactions Ordinance and related PTA and NCCIA notifications from the Pakistan Code, the Gazette of Pakistan and official websites [1]–[4]; reading text from PDFs (OCR for scanned pages); and splitting each document into sections and clauses with its number, department, publication date and effective date.

3.1 Regulatory Ontology Design

The ontology has a general core that works for any area of law, and a cyber law extension. It will be built in OWL 2 using Protégé, use the W3C OWL-Time and PROV-O vocabularies for dates and sources, and be tested with a reasoner and our list of questions (Table 1).

Concept	General Meaning	Cyber Law Example
Law / Regulation	An Act, Ordinance or set of Rules; *contains* provisions.	PECA 2016; Content Blocking Rules 2021.
Notification / Amendment	*Inserts*, *substitutes*, *omits* or *repeals* a provision; has an effective date.	PECA (Amendment) Act 2025 inserting section 26A.
Provision / RuleVersion	A section and each dated version of its text; *replaces* the earlier version.	Section 26A, valid from 29 January 2025.
Requirement / Obligation	A condition, document, fee or duty that *appliesTo* a service or person.	Duty of platforms to remove unlawful content.
Offence / Penalty	Conduct *punishableBy* a penalty (cyber law extension).	False information: up to 3 years and/or Rs. 2 million.
Department	A body that *issues* rules and *provides* services.	PTA, NCCIA, Social Media Protection and Regulatory Authority.
Government Service	A public service that *hasRequirement* requirements.	Cybercrime complaint; content removal request.
TimePeriod / Source	When a fact is valid, and the document and clause it comes from.	Gazette of Pakistan, 29 January 2025.

Table 1: Main ontology concepts with cyber law examples

3.2 Temporal Knowledge Graph Design and Construction

The ontology will be stored as a graph in Neo4j. As in [12], each *Provision* has one fixed node, and every change to its text adds a new *RuleVersion* node linked by *HAS_VERSION* and *SUPERSEDES* edges; an *Amendment* node links the amending document to the old and new versions, so no history is lost. Each version keeps two kinds of dates [11]: *valid_from / valid_to* (when it was in force) and *published_on / added_on* (when it was announced and added). Links such as *INVESTIGATED_BY*, *PUNISHABLE_BY* and *HAS_REQUIREMENT* also carry dates, because, for example, cybercrime investigation moved from the FIA Cyber Crime Wing to NCCIA without the offences themselves changing. The agency investigating an offence on date *t* can be found with:

```
MATCH (o:Offence {name:$offence})-[r:INVESTIGATED_BY]->(d:Department)
WHERE r.valid_from <= date($t) AND (r.valid_to IS NULL OR r.valid_to > date($t))
RETURN d.name, r.source_doc, r.clause_ref
```

To fill the graph, text patterns first find section numbers, dates and amendment words (“inserted”, “substituted”, “omitted”); an LLM then extracts entities and relationships in an ontology-based JSON format; and each fact is checked, linked to its source clause and reviewed by us if confidence is low.

3.3 Temporal Graph-RAG Question Answering Pipeline

Normal RAG only searches for similar text. In our pipeline, the graph first decides *which* rule was valid on the asked date, and the search then finds *where* it is written:

1. **Understanding the question:** the LLM finds what the user wants (change, applicable version or impact), the rule, offence or service mentioned (e.g., "fake news" → PECA section 26A) and the date (e.g., "in 2024"; otherwise today).
2. **Searching the graph:** ready-made Cypher queries (with LLM-written ones as backup) collect the rule versions, offences, penalties, requirements, services and departments valid on that date.
3. **Searching the text:** clauses are searched by keywords (BM25) and by meaning (embeddings); only clauses valid on that date are kept, the results are merged, and a re-ranking model puts the best matches on top.
4. **Preparing the context:** graph facts and best clauses form a prompt that limits the LLM to this evidence.
5. **Answering and checking:** the LLM answers with citations (document, section, effective date); each citation is checked for date validity, and if evidence is missing the system says so.

3.4 Change Detection, Version Resolution and Impact Analysis

- **Change detection:** for a new notification, the system finds the amended rule and its effective date, compares the old and new versions clause by clause, and labels each change (e.g., the 2025 amendment *inserting* section 26A).
- **Finding the applicable version:** for a date *t*, the version with valid_from ≤ t < valid_to is selected, and unclear overlaps are flagged.
- **Impact analysis:** from a changed rule, the system follows the graph links to list the affected requirements, services, offences, departments and people, e.g., showing that the 2025 amendment affects complaint handling, platform duties and the investigating agency.

3.5 Technology Stack and Resources

Category	Technology / Tool	Purpose
Ontology Engineering	OWL 2, Protégé, HermiT reasoner	Building and checking the ontology
Graph & Retrieval	Neo4j (Cypher, vector index), BM25, embeddings, re-ranker	Dated graph storage, keyword and meaning-based search
Large Language Model	GPT-4o mini (API) or Llama 3.1 via Ollama	Extracting facts, understanding questions and writing answers
Application	Python, FastAPI, LangChain, PyMuPDF, Tesseract OCR, React.js, Docker	Pipelines, OCR, APIs, web interface and deployment

Table 2: Technology stack and resources

3.6 Testing and Evaluation

We will prepare test questions on PECA and related rules with dates and correct references (including dates before and after the 2025 amendment), and compare our system with an LLM alone, normal RAG and Graph-RAG without dates, using:

- **Extraction and search:** precision, recall and F1-score; Recall@k and MRR for date-valid clauses [15].
- **Date and impact accuracy:** correct version, change type and affected items (e.g., FIA vs. NCCIA by date).
- **Answer quality:** RAGAS faithfulness and context precision [17], citation accuracy, made-up answer rate and response time, plus unit, integration and system testing.

4. Project Scope

Included Functionalities:

- A general regulatory ontology with a cyber law extension, and a dated knowledge graph of PECA 2016, its amendments and related rules.
- Change detection, applicable version finding and impact analysis on services, offences and departments.
- Date-aware Graph-RAG question answering on cyber law in English, with references, in a web interface.
- Court judgments as change events, limited to the 2022 Islamabad High Court judgment that struck down the PECA (Amendment) Ordinance, 2022 and part of section 20 [19].

Excluded Functionalities:

- Other legal domains (planned as future extension), and court judgments and case law other than the selected 2022 judgment.
- Legal advice and live tracking of the Gazette.
- Urdu support, mobile apps and government system integration.

5. High level Project Plan

The project will run from October 2026 to June 2027, as shown in Figure 1. Yumna Aziz (Team Lead) will lead data collection, the ontology and the knowledge graph, including change detection, version finding and impact analysis; Hamna Tariq will lead the LLM components (extraction and question answering), the Graph-RAG pipeline, the web interface and evaluation. Both will share annotation and writing under the supervision of Dr. Muhammad Ilyas, Dr. Muhammad Saad Razzaq and Ms. Sabahat Sabir.

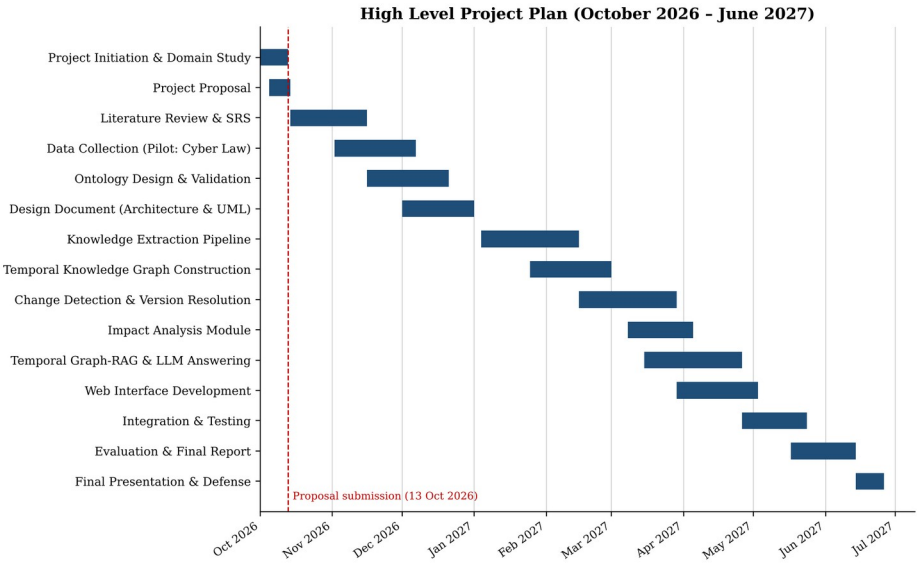


Figure 1: Gantt chart of the high-level project plan

Key Milestones:

Milestone / Deliverable	Planned Date
Project Proposal submission	13 October 2026
Software Requirements Specification (SRS)	15 November 2026
Regulatory ontology and Design Document	31 December 2026
Prototype: extraction pipeline and temporal Knowledge Graph	28 February 2027
Change detection, version resolution and impact analysis	04 April 2027
Integrated system with temporal Graph-RAG and web interface	23 May 2027
Final report, presentation and defense	June 2027

Table 3: Key milestones

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